S1 Text. Supplementary methods and verification. Implementation equivalences and an end-gap check, the optional recursive partial alignment, parameter sensitivity, CIRR-Hybrid anchoring controls, SRSC parameter selection, the MIA reruns and ambiguous-base verification, a circular-boundary diagnostic, control-region diagnostics, read-placement and mixed-orientation checks, and the conditional accuracy of the *de novo* assemblers. Lettered tables and algorithms belong to this file; numbered tables, figures, equations, and algorithms refer to the main text.

1 Current benchmark provenance and scope of the historical controls

The current main paper reports the accepted 30-species panel: 870 ordered pairs, eight methods, depths 1–8X, three technical replicates, and 167,040 result rows. Main-grid source is the frozen benchmark commit `226a2892a6dd481deb1e6539ffc97320e998f690`; the SHA-256 of its source/input manifest is `ad97b480a376 38304ae12ef9139928e569bf538e9e73f7b767b3bbcd3189bb07`. MITObim uses inputs exported from that same manifest and the frozen neutral-ambiguity evaluator. Output hashes, exact key coverage and metric-formula consistency passed acceptance for the main grid and the two follow-ups. The main-grid method keys `ASEM` and `ASEM_Hybrid` mean CIRR and CIRR-Hybrid throughout the machine-readable files.

The 12-pair mixed-orientation and higher-depth experiments use a separately hashed, strand-aware patch snapshot dated 1 November 2026. Its forward controls match the main-grid F1 and identity on all 288 corresponding CIRR/Hybrid jobs. Mixed and forward controls are sequence-identical in all 288 comparisons. This tests the revised implementation on the selected pairs. Main-grid forward-only scores remain separate. MITObim was run on the full main grid and has no matched follow-up, empirical, read-length or controlled-runtime rows.

S2 Table is refreshed for this panel and contains the job-level main scores, conditional summaries, paired differences, species deletions, unchanged-reference baselines, net aligned-count changes, and follow-up summaries. LOSO min/max values describe sensitivity to deleting each observed species in both dyad roles; they are not confidence intervals. Unchanged-reference differences compare the matching fixed or normalized evaluation policy. Net matched-base changes do not locate individual corrected sites.

The sections below preserve earlier six-species and targeted mechanistic controls. Their 720-job counts, original mixed-strand loss, earlier convergence counts, and legacy assembly-depth results refer to those stated experiments, not to the new main grid. S3 and S4 Tables retain the earlier MIA retry and evaluator-correction histories for the same reason. The current orientation result supersedes the historical orientation limitation for the tested strand-aware follow-up. S5 and S6 Tables use the latest internal tie-rule reruns and the retained neutral-evaluator external rows, with source hashes in the manuscript provenance. The published archives cited in the main paper predate this refresh; a new public release remains pending.

2 Implementation equivalences

The historical diagnostics in the following sections used the open Python 3 benchmark implementation (archived benchmark v1.3.0), in whose code and result files CIRR, CIRR-Hybrid, and recursive CIRR appear under their working names `ASEM`, `ASEM_Hybrid`, and `ASEM_recursive`. Three dependencies of earlier work that are unavailable or unmaintained were replaced.

Local alignment. Read alignment uses `sw_trace_striped_16` from `parasail` (Daily 2016), which implements Farrar's striped Smith–Waterman algorithm (Farrar 2007), the algorithm of the former scikit-bio `StripedSmithWaterman` class used in the original work. For the read and reference sizes in this study, `parasail` was about 15 times faster than scikit-bio's current general-purpose aligner.

Read simulation. A custom single-genome, uniform-random shotgun simulator reproduces the default single-reference shotgun behavior of Grinder 0.5.4 (Angly et al. 2012), which was used in the original work but is no longer distributed through Bioconda. Seeds are `zlib.crc32` hashes of the identifying tuple, not Python's process-randomized `hash()`.

Evaluation. scikit-bio's `pair_align_nucl` in global mode with its default `free_ends=True` gives an overlap (semi-global) alignment with unpenalized terminal gaps. It is therefore not a strict reproduction of EMBOSS `needle`, which the original work used; full quadratic-space `needle` required 30–90 s and about 4 GB per 16.5-kb pair, which was impractical for the benchmark grids. As an end-gap sanity check, evaluation was repeated with the same dynamic program but `free_ends=False` (strict end-to-end global alignment) for one 4X replicate of a representative pair at each divergence level. Results were identical for the same-genus and same-family jobs; for the same-order job, strict-global evaluation increased F1 by 0.00246 (0.91960 to 0.92206) and identity by 0.552 percentage points. Unpenalized ends therefore did not inflate the reported values in these jobs; this three-job check is not a full evaluator cross-validation.

3 Recursive partial alignment

Specification. A read may be broadly similar to the reference yet contain a segment with no counterpart; local alignment then stops short of the read ends. Recursive partial alignment realigns each unaligned flank at least $\ell = 30$ bases long to $\Theta^{(t)}$ independently, and recurses whenever a flank itself leaves a long enough unaligned remainder (Algorithm A). Recovered fragments enter the combined alignment as additional rows.

Algorithm A Per-read local alignment with recursive retry

Require: Read r, current sequence Θ
Ensure: Set L of accepted local alignments (aln, pos)
  procedure AlignRead(r, Θ)
    $L \leftarrow \{\}$
    $(score, aln, pos) \leftarrow \text{LocalAlign}(r, \Theta)$
    if $score/(s_{\text{match}}|r|) \geq \tau$ **then**
      $L \leftarrow L \cup \{(aln, pos)\}$
    end if
    $(prefix, suffix) \leftarrow \text{UnalignedFlanks}(r, aln, pos)$
    for each $r' \in \{prefix, suffix\}$ **do**
      if $|r'| \geq \ell$ **then**
        $L \leftarrow L \cup \text{AlignRead}(r', \Theta)$
      end if
    end for
    return L
  end procedure

Results. On the earlier six-species grid, recursive and non-recursive CIRR differed in at least one reported final metric or sequence length in 53 of 720 jobs (33 same-order, 20 same-family). F1 differed in 36 jobs, with a mean absolute difference of 0.00015 among them and a signed mean difference below 0.000003 over all 720 jobs; divergence-level means agreed to the reported precision (same-family F1 0.9777 versus 0.9776). In the read-length experiment, F1 differed in 2 of 63 jobs, with a maximum absolute difference of 0.00011. On the empirical NA07000 reads, F1 differed in 1 of 24 chimpanzee-reference and 5 of 24 *Varecia*-reference jobs, by at most 0.00024. Run time and memory were similar to non-recursive CIRR in the controlled benchmark (median 0.4–5.6 s per job at 1–8X, peak 172 MB). Recursion is designed for reads split across a long structural breakpoint, which the benchmark did not include; within the tested data it had no practical effect, and the main text therefore reports non-recursive CIRR.

Mechanism. In the hardest pair (same-order, 8X, *Homo sapiens*/*Varecia variegata*), none of the 905 simulated reads triggered an accepted second alignment. Even reads in the D-loop, where local mismatch approached 100%, aligned for 149 of 150 bases in a single Smith–Waterman call, because conserved positions accumulated enough score to extend the alignment through scattered mismatches, leaving no remainder long enough to trigger recursion. In a synthetic translocation, a 1,200-bp segment of *Saimiri boliviensis* mtDNA was excised and reinserted 7,000 bp away, and 1,648 reads simulated from the rearranged target were aligned

to the unmodified reference. A single alignment left a remainder of at least 30 bp in 33 reads (about 2.0%), close to the about 1.7% expected from three junctions ($3 \times 91/16{,}471$), and recursive alignment recovered an accepted second alignment for all 33. Recursion can therefore recover fragments across a long structural breakpoint, but the linear consensus still cannot represent the rearrangement itself.

4 Parameter sensitivity

Design. The alignment threshold was varied over $\tau \in \{0.30, 0.40, 0.50, 0.60, 0.70\}$, the minimum-support fraction over $w \in \{0.05, 0.10, 0.20, 0.30\}$, and the iteration cap over $t_{\max} \in \{1, 3, 6, 10\}$, one factor at a time with the others at their defaults, on one representative pair per divergence level at 1, 2, 4, and 8X with three replicates (396 jobs). Target truth was never available to the running algorithm.

Results. Across all divergence levels, mean F1 was 0.9572 at the default $w = 0.10$, 0.9572 at $w = 0.05$, 0.9567 at $w = 0.20$, and 0.9561 at $w = 0.30$. Raising $t_{\max}$ from 1 to 3, 6, and 10 gave mean F1 of 0.9458, 0.9548, 0.9572, and 0.9573. The gain from six to ten iterations was 0.00015 overall and 0.00046 for the same-order pair; 25% of default jobs reached the six-iteration cap, against 2.8% with $t_{\max} = 10$. For the same-order pair, mean F1 decreased from 0.9408 at $\tau = 0.30$ through 0.9244, 0.9001, and 0.8552 to 0.8422 at $\tau = 0.70$; the range was 0.9718–0.9674 for the same-family pair and 0.9999 throughout for the same-genus pair. A permissive threshold therefore helped this clean, single-target simulation. We retained the prespecified $\tau = 0.50$, because choosing it on known target truth would optimize the benchmark post hoc, and the simulation omits contaminants, nuclear mitochondrial segments, and repeats that raise the cost of false placements.

5 CIRR-Hybrid anchoring controls

With $\tau_c = 0.25$, $s_{\text{match}} = 2$, and mismatch penalty -3, the contig acceptance criterion corresponds to at least 70% identity in the ungapped case ($2p - 3(1-p) \geq 0.5 \Leftrightarrow p \geq 0.7$). The more permissive per-base threshold than for reads is justified by length: the probability that unrelated sequences keep a given identity by chance decreases with alignment length. We generated 2,000 independent uniform-ACGT sequences at each of 150 and 300 bp, unrelated to the reference, and required an aligned length equal to the sequence length rather than the default 300 bp; no false anchor was accepted at either length. This null model does not represent GC bias, repeats, or low-complexity tracts, which may increase spurious anchoring, and it does not test the distinct risk that two unrelated reads share an exact 20-bp suffix–prefix overlap and are merged into a chimeric contig before anchoring. The current safeguards are flank-based local read selection, a 20-bp exact overlap, and at least 300 bp of anchored alignment at the 70%-identity-equivalent score. A low-complexity mask should be evaluated on structured genomic negative controls before it is added.

6 SRSC parameter selection

The minimum match fraction of SRSC, which controls the precision–recall trade-off in read anchoring, was not disclosed in the original paper; we used 0.25 after tuning on the most difficult same-order pair. The original paper also did not specify the overlap between adjacent chunks, which determines the flank used for merging. An untuned 100-bp overlap caused a pronounced F1 decrease at read lengths of about 100–250 bp and an F1 decline of about 0.06 from 1X to 8X on same-order pairs. A post-hoc experiment that varied only the overlap on identical reads showed that 150 bp substantially reduced, but did not eliminate, the read-length decrease across 50–300 bp: in the three-replicate read-length experiment, one 250-bp replicate reached F1 0.729 while the other two were about 0.935, and at 300 bp two of three replicates fell to 0.805. At 150 bp, the 30X results were also closer to the originally reported values (same-family recall 0.60→0.95 and identity 55.0%→86.7%; same-order recall 0.79→0.91 and identity 61.7%→70.3%). The 150-bp overlap remains a post-hoc choice within the tested range, not a generally validated value.

7 MIA reruns and ambiguous-base verification

Time limit. In the grid runs, MIA jobs had a 300-s limit. Every job that reached it or ended with a process error was rerun once with the original inputs, flags, and evaluator under a 3,600-s limit, with full logs retained. The 69 jobs that had reached the 300-s limit had not stalled: 68 finished in 50–314 s (median 92 s) on an unloaded machine. No rerun reached the 3,600-s limit.

Scoring correction. For the neutral scoring of ambiguous bases, every MIA job of the earlier six-species grid, every MIA, Pilon, NOVOPlasty, and GetOrganelle job of the *Macaca* experiment, and every MIA, Pilon, and NOVOPlasty job of the read-length and empirical-read experiments (GetOrganelle returned no sequence in the latter two) was rerun with output sequences retained, and each output containing ambiguous bases was rescored with both scoring rules (S4 Table). The earlier-grid MIA rerun ran nine jobs at a time; every job that failed was retried once alone on the machine, and a retry that returned a sequence replaced the failure. Reported MIA values come from these reruns.

Under the original evaluator, rerun outputs matched the recorded F1 in 692 of the 700 earlier-grid jobs that completed in both runs. Neutral N scoring left F1 unchanged in 360 of these jobs and changed it in 332; 199 of the latter had been scored F1= 0 and now scored 0.38–0.75. The original runs did not retain their output sequences, so matching F1 is consistent with, but does not prove, an identical sequence. In the remaining 8 jobs, all *Homo sapiens*→*Gorilla gorilla* at 1–3X, the original-evaluator F1 differed from the recorded value by at most 0.0091; two of them, run twice more, reproduced the rerun sequence exactly (identical SHA-256 checksums). MIA's process errors were intermittent: 3 of the 21 jobs that failed in the parallel rerun returned a sequence on retry, 2 of which had also failed in the original grid and its follow-up, and 1 job that had completed in the original grid failed in both reruns. The final earlier-grid outcome is 702 of 720 jobs with output; the 18 failures (12 aborts and 6 segmentation faults) all involve same-order pairs, 16 of them with *Varecia variegata*.

Pilon outputs contained at most one ambiguous base, and Pilon's F1 equaled the recorded value in all 159 rerun Pilon jobs, so its earlier-grid jobs were not rerun. Main-grid NOVOPlasty and GetOrganelle outputs, including the 10–30X extension, had been retained and were rescored directly: ambiguous bases occurred in 5 NOVOPlasty outputs (4 at 15X and 1 at 7X; F1 change at most 0.0001) and in no GetOrganelle output.

8 Circular-boundary diagnostic

CIRR stores the circular mitochondrial genome as a linear string, and the simulated reads were sampled from the linear target, so no read spanned the origin. To test whether origin-spanning reads would leave the sequence ends with less support, a human mtDNA target was paired with three synthetic starting references carrying uniformly placed substitutions at 5%, 10%, or 20%. For each divergence, depth (1, 2, 4, 8X), and replicate (10 per cell), read sets were generated with the same seed by linear sampling or by circular sampling, which lets a read wrap across the origin; CIRR itself was unchanged. Across 120 matched comparisons, circular sampling increased mean accuracy in the first and last 150 target bases by 1.89 percentage points (range over runs 0 to 14.67) and changed interior accuracy by -0.027 percentage points; the terminal difference was non-negative in every cell. Local alignment thus recovered one informative segment of an origin-spanning read, and the linear-sampling benchmark was mildly conservative at the terminal coordinates. This substitution-only diagnostic does not establish circular-genome support for indels, rearrangements, or CIRR-Hybrid's overlap assembly.

9 Control-region diagnostics

At 8X, CIRR's same-family outputs had one to seven mismatched bases per genome but lacked 479–680 target bases, all in the control region. When both genomes were rotated so that the control region lay mid-sequence, CIRR left as much of it unrecovered (579–762 target bases at 8X), so the result is not an artifact of the linear origin. Neither ten iterations nor a permissive placement threshold ($\tau = 0.3$) changed the same-family result in the sensitivity analysis, and CIRR-Hybrid did not either, because the region was not

completely without placed reads. MIA recovered the region in one to four iterations in its distant-reference mode.

10 Read placement and orientation checks

Multiple placements. Each read contributes only its best placement. In the hardest pair (same-order, 8X), the selected position of each read was masked and the best alternative alignment recomputed. None of 905 reads had a secondary placement that passed the alignment threshold; 45 had a secondary score at least 90% of the primary, but both scores were only 6–12% of the theoretical maximum. The other two replicates and the final refined reference gave the same result (0/905 in every case).

Mixed read orientation. All simulated grid reads were forward-strand. The hardest same-order pair (*Homo sapiens* reference, *Varecia variegata* target) was rerun at 1, 2, 4, and 8X with three replicates using the archived reference implementation v0.4.1, once with the grid reads and once after reverse-complementing each read with probability 0.5; target truth was not used to reorient reads. Ordinary CIRR, whose aligner tests both strands, returned identical F1 in all 12 matched jobs. CIRR-Hybrid lost F1 at low depth (mean change −0.0074 at 1X, −0.0184 at 2X, −0.0048 at 4X, and 0 at 8X; largest single-job loss 0.025), consistent with opposite-strand reads failing to merge in its forward-only overlap assembly, but remained above ordinary CIRR at every depth (0.856 versus 0.852 at 1X, 0.896 versus 0.872 at 2X, 0.951 versus 0.906 at 4X, and 0.964 versus 0.928 at 8X). These package-level values come from a separate implementation of the same procedure and are compared only with each other.

11 Conditional accuracy of the de novo assemblers

Table A: Conditional performance of the *de novo* assemblers on the earlier six-species grid

	NOVOPlasty		GetOrganelle	
Depth	F1 (successful/total)	Identity (%)	F1 (successful/total)	Identity (%)
1–8X	0.134 (274/720)	7.4	0.728 (3/720)	57.3
10X	0.365 (87/90)	24.0	0.887 (10/90)	83.1
15X	0.828 (79/90)	78.3	0.893 (41/90)	82.8
20X	0.865 (77/90)	83.9	0.891 (63/90)	81.5
30X	0.861 (74/90)	83.4	0.972 (81/90)	95.1

Mean F1 and identity among successful jobs; failures were not assigned F1= 0. Both assemblers used the same 30 ordered pairs and three replicates. Strand and circular-origin normalization changed F1 in 67 of 198 successful GetOrganelle jobs (by up to 1.0, consistent with correctly assembled scaffolds that were reverse-oriented or rotated) and in 3 of 591 successful NOVOPlasty jobs (by at most 0.046). On the easiest same-genus pairs, NOVOPlasty assembled 32 of 48 jobs at 1–8X, with success-only mean F1 of 0.10. On the verified *Macaca* pair, GetOrganelle produced a scaffold in 1 of 48 jobs at 1–8X, 3 of 6 at 10X, and every job from 15X onward, with success-only F1 of at least 0.999; NOVOPlasty returned a sequence in 23 of 48 jobs at 1–8X with success-only F1 no higher than 0.19, and F1 of at least 0.998 in every job from 15X onward.

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Supplementary file guide

S1 Text. Supplementary methods and verification. Implementation equivalences and an end-gap check, the optional recursive partial alignment and its results, parameter sensitivity, CIRR-Hybrid anchoring controls, SRSC parameter selection, the MIA reruns and ambiguous-base verification, a circular-boundary diagnostic, control-region diagnostics, read-placement and mixed-orientation checks, and the conditional accuracy of the *de novo* assemblers.

S1 Fig. Complete pair-level heatmaps at 8X. Eight pages, one method per page, with all 30 references and targets labelled. Colors show mean F1 over scoreable replicates on a common 0–1 scale. Gray crosses denote no output; the blank diagonal was not attempted. Digits 1 or 2 mark cells with fewer than three outputs. The main paper retains the tier/depth summaries and output-rate curves.

S1 Table. Mitochondrial genomes used in the experiments. Species, family, order, NCBI RefSeq accession, and use of every mitochondrial genome in the study.

S2 Table. Accepted 30-species grid and post-processing. Eight methods, 167,040 job-level scores and statuses, conditional summaries by tier and depth, unchanged-reference scores and net changes, paired method comparisons with species deletion sensitivity, each species deletion, and orientation/high-depth follow-up summaries. Paired differences are anchor minus comparator, with the anchor identified on every row. LOSO ranges are sensitivity ranges, not confidence intervals. Raw method keys retain the historical code names. The read-me sheet defines missing outputs and the secondary failure-zero utility. The preceding six-species S2 workbook is retained in the pre-refresh provenance archive.

S3 Table. Job-level outcome of every MIA run. For the earlier 720-job six-species MIA grid (not the refreshed panel): original grid status and wall time, follow-up status and wall time under the 3,600-s limit, final status after the scoring-correction rerun and the sequential retry of failed jobs, and the reported F1, identity, recall, and precision with the checksum of the stored output sequence.

S4 Table. Historical rescoring of external-tool outputs with the neutral N score. For every rerun MIA, Pilon, NOVOPlasty, and GetOrganelle job: number of ambiguous bases in the output, F1 recorded originally, F1 of the rerun output under the original and the neutral N scoring, and whether the original-scoring F1 matched the recorded value (the original outputs were not retained, so sequences themselves could not be compared). Rerun failures that returned a sequence on the sequential retry are verified through the retry output. The rescoring of stored main-grid NOVOPlasty outputs and a repeated-run check of MIA output are summarized in S1 Text.

S5 Table. Read-length experiment, seven measured methods. Job-level status, run time, F1, identity, precision, and recall for every method, pair, read-length condition, and replicate at 4X. CIRR/Hybrid rows use the latest internal tie-rule rerun; MIA, Pilon, and NOVOPlasty rows use the retained-output neutral-evaluator rerun. MITObim is not measured. Cell means are in Table 6.

S6 Table. Empirical-read experiment (NA07000), seven measured methods. Job-level status and F1, identity, precision, and recall against the phase-3 truth with the primary mask, the strict mask, and no mask, and against rCRS, for every method, starting reference, depth 1–8X, and replicate. CIRR/Hybrid rows use the latest internal tie-rule rerun; MIA, Pilon, and NOVOPlasty rows use the retained-output neutral-evaluator rerun. MITObim is not measured. Depth means are in Table 7.